

Using Realtime GTFS to generate easy-to-use transit accessibility measures under travel time uncertainty[☆]

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ABSTRACT

Previous studies on measuring transit accessibility under travel time uncertainty often introduced complex measures based on non-standard data formats, hindering reproducibility and replicability in research and planning. To address this, we present a practical framework that leverages a standardized format for real-time transit data: Realtime General Transit Feed Specification (GTFS), to generate easy-to-use transit accessibility measures under travel time uncertainty. This framework first produces two datasets by correcting Scheduled GTFS data using Realtime GTFS information: Realtime P50 GTFS and Realtime P85 GTFS, which are used to compute two accessibility measures: *median-corrected accessibility* (using Realtime P50 GTFS) and *dispersion-corrected accessibility* (using Realtime P85 GTFS). These accessibility measures are applied in Columbus, Ohio, USA for an empirical study examining how overlooking travel time uncertainty issues can distort the analysis results of healthcare accessibility, inequality, and new transit project evaluation. Results indicate that *scheduled accessibility* (using Scheduled GTFS data) which overlooks travel time uncertainty overestimates healthcare accessibility by approximately 10.97 %. Moreover, this oversight fails to capture the benefits of the new transit service in improving accessibility and reducing inequality. Furthermore, although findings consistently suggest that lower-income neighbourhoods experience greater gains in healthcare accessibility compared to wealthier counterparts, our analysis unveils statistically significant differences when using the scheduled and dispersion-corrected accessibility measures. These findings underscore the importance of incorporating travel time uncertainty into public transit planning and evaluation. Our framework allows transit authorities and researchers to accurately measure accessibility and evaluate projects under travel time uncertainty using a standardized data format.

1. Introduction

Measures of transit-based accessibility are useful for evaluating the performance of land-use and transportation systems, informing policy-makers in decision-making, and addressing transportation issues such as unequal accessibility (Geurs and van Wee, 2004; Levinson and King, 2020; Siddiq and Taylor, 2021). Traditionally, most studies on public transit accessibility rely on Scheduled (or Static) General Transit Feed Specification (GTFS) data to compute transit travel times and accessibility metrics (Ahmed et al., 2024; Javanmard et al., 2024; Klar et al., 2023; Lee and Miller, 2018, 2019; Pereira, 2019; Singh et al., 2022; Manaugh and El-Geneidy, 2012). This approach often assumes perfect adherence to schedules without any delays or early arrivals, an

assumption frequently challenged by real-world conditions (Chen et al., 2013; Lee and Kim, 2023; Liu et al., 2022b; Wessel et al., 2017; Wessel and Farber, 2019). However, this assumption can result in erroneous measurements of transit accessibility and evaluations of transit interventions.

To address the limitations of *scheduled accessibility* measures (using Scheduled GTFS data) in transit-based accessibility analysis, recent literature has shifted towards incorporating travel time uncertainty to achieve more accurate and realistic measurements of accessibility via public transit. While this shift improves overall transit accessibility evaluation, it is particularly critical for healthcare access, where even minor delays can result in missed or rescheduled appointments. Many healthcare facilities enforce strict policies, requiring patients to wait

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weeks for a new slot or pay cancellation fees, which disproportionately burden low-income individuals with inflexible work schedules or limited transportation options. This can create a double disadvantage, restricting access to care while adding financial strain. Therefore, addressing travel time uncertainty is essential for ensuring equitable accessibility to critical destinations such as healthcare, since more accurate and realistic accessibility measures can inform planning and policy efforts to mitigate these barriers for transit-dependent populations.

Recognizing these challenges, researchers have sought to refine accessibility measurements by using GPS or Automatic Vehicle Location (AVL) data from buses to account for travel time uncertainty. For example, [Zhang et al. \(2018\)](#) developed a multimodal transit-based accessibility model that uses GPS data to estimate travel time distributions between bus stops. Their research demonstrates that this time-dependent reliability modeling approach provides more accurate insights into temporal variations in transit accessibility. Also, [Wessel and Farber \(2019\)](#) compared traditional accessibility measures derived from Scheduled GTFS data with actual accessibility from archived GPS-based AVL data from four North American transit agencies. Their study revealed that scheduled accessibility measures (using Scheduled GTFS data) may overestimate actual accessibility by about 5–15 percent or more. Most recently, [Braga et al. \(2023\)](#) addressed the impacts of travel time uncertainty on transit accessibility and inequality in Fortaleza, Brazil, by correcting Scheduled GTFS feeds using historical bus GPS data. They found that Scheduled GTFS data often overestimates job accessibility, but sometimes underestimates it in specific areas, highlighting the complex challenge associated with measuring transit accessibility.

Despite this progress, reliance on historical GPS or AVL data in custom formats is limited by its reproducibility and replicability in transportation research and practice. Reproducibility is hindered because these datasets often require extensive preprocessing and custom adjustments to align with different transit systems, study objectives, and analytical frameworks. Since preprocessing steps are often not standardized or well-documented, minor differences in data cleaning, aggregation, or filtering can lead to variation in results, making it difficult for other researchers or practitioners to reproduce findings using the same dataset. Replicability is further complicated by the inconsistent availability and format of GPS and AVL data across different studies and regions. When data is collected and stored in proprietary or non-uniform formats, replicating analyses requires significant customization and often involves case-by-case data adaptation. This lack of standardization limits the applicability of methodologies across multiple systems, reducing comparability and generalizability in research and policy evaluation.

In contrast, Realtime GTFS data, a *standardized* format for providing real-time transit operations data globally ([Google, 2024a](#)), addresses these limitations and promotes reproducibility and replicability. Several studies have used Realtime GTFS data to evaluate the impacts of travel time uncertainty on transit accessibility. For example, [Lee and Miller \(2020\)](#) used Realtime GTFS data to develop a robust space–time accessibility (RSTA) methodology, which considers various risk management strategies to measure accessibility under travel time uncertainty. [Lee and Kim \(2023\)](#) further employed the RSTA approach alongside the Gini coefficient and spatial regression, to demonstrate how scheduled accessibility measures, which rely on Scheduled GTFS data, fail to capture significant disparities in transit-based healthcare accessibility. Additionally, [Liu et al. \(2022a\)](#) compared traditional accessibility measures with realizable accessibility measures derived from Realtime GTFS data, by employing space–time prisms concepts. Their results highlight the overestimation of accessibility by scheduled measures, which do not account for travel time uncertainty. While these studies effectively represent actual transit accessibility under travel time uncertainty using standardized transit data formats, they are complex and computationally intensive. This complexity necessitates significant data

processing power, specialized technical expertise, and intricate calculations, which are often inaccessible to smaller transit agencies or researchers without a strong technical background. As a result, transport planners and researchers may struggle to adopt these methods, limiting their practical and research applications.

To fill this gap, we introduce a practical framework that uses a standardized format for real-time transit data: Realtime GTFS data, to compute easy-to-use transit accessibility measures under travel time uncertainty. A key innovation of this study is its ability to tackle the reproducibility and replicability challenges of studies relying on non-standardized GPS or AVL data, as well as the complexity challenges of previous Realtime GTFS-based studies. Our approach simplifies and clarifies the integration of real-time transit data into accessibility analysis, making it more user-friendly to transit authorities and transport researchers with limited technical expertise or resources. The framework first creates two datasets by correcting Scheduled GTFS feeds based on Realtime GTFS information: the median-corrected dataset (Realtime P50 GTFS) and the dispersion-corrected dataset (Realtime P85 GTFS). For the Realtime P50 GTFS, we use the median (50th percentile) travel time to correct Scheduled GTFS. The median value represents the *typical* travel time experienced by passengers. For the Realtime P85 GTFS, we use the 85th percentile travel time, which captures a more *conservative* estimate of travel time that accounts for the upper range of variability. Based on this, we then compute two cumulative-opportunity measures under travel time uncertainty proposed by [Braga et al. \(2023\)](#): *median-corrected accessibility* (using Realtime P50 GTFS) and *dispersion-corrected accessibility* (using Realtime P85 GTFS). Unlike previous approaches relying on custom formats and computationally demanding techniques, these accessibility measures are easy to use because corrected GTFS datasets can be directly imported into commonly used transportation analytics tools such as *r5r* ([Pereira et al., 2021](#)) and ArcGIS Pro Public Transit toolbox ([Esri, 2024](#)) for generating median-corrected and dispersion-corrected accessibility metrics. We elaborate and apply this approach in an empirical study exploring how overlooking travel time uncertainty issues can distort the assessment of a new public transit service's impacts on healthcare accessibility and inequalities in Columbus, Ohio, USA. We also examine how these impacts vary spatially and across income groups. Ultimately, this study aims to provide transportation scientists and practitioners with a more realistic, practical, and easy-to-use approach based on a standardized data format to accurately measure transit accessibility and evaluate the accessibility impacts of transit projects under travel time uncertainty.

The remainder of the paper is organized as follows: [Section 2](#) details our methodology, providing an overview of the steps taken to correct Scheduled GTFS using Realtime GTFS feeds and generate accessibility measures. Following this, [Section 3](#) presents an empirical case study examining the impacts of travel time uncertainty on healthcare accessibility, inequality, and transit project evaluation in Columbus, Ohio, USA. Finally, [Section 4](#) discusses the implications and limitations of our research, along with suggested areas for future studies.

2. Methods

2.1. Correcting scheduled GTFS feeds using Realtime GTFS data

We begin by computing the daily travel times for links between two consecutive stops for each trip during the study period (e.g., one month, summer). A trip in GTFS data is defined as a series of two or more stops along a route within a designated timeframe ([Google, 2024b](#)). A link, on the other hand, refers to the connection between two consecutive stops within a trip. Following this, we create a travel time distribution for each link between stops over the study period. By compiling these travel times, we generate a travel time distribution that reflects the variability in travel times for each link. From these distributions, we calculate specific percentiles: the median (50th percentile) and the 85th

percentile travel times for each link between stops.

We adopt the median and 85th percentile travel times because day-to-day travel time distributions are often skewed, making traditional mean and variance less representative of actual travel time distributions (Van Lint and van Zuylen, 2005). The median provides a robust estimate of typical travel times under usual conditions, indicating that commuters are likely to reach their destination within this timeframe at least 50 % of the time (Chen et al., 2017). This measure is pertinent for activities with flexible scheduling such as leisure trips or shopping where exact arrival times are less critical (Conway et al., 2018). However, for critical appointments where punctuality is essential, such as job interviews, flights, or medical visits, the 85th percentile offers a more conservative measure. It reflects longer travel times that accommodate usual delays and ensure reliability in 85 % of cases (Conway et al., 2018).

We then match these calculated travel times with their corresponding links in the Scheduled GTFS *stop_times.txt* file. This step involves aligning the real-time travel data with the scheduled times to identify discrepancies and apply corrections. In the next step, we replace the scheduled departure and arrival times in the *stop_times.txt* file with the corrected travel times. Specifically, for the Realtime P50 GTFS dataset, the median travel times replace the scheduled times, creating a newly corrected GTFS dataset. Similarly, for the Realtime P85 GTFS dataset, the 85th percentile travel times replace the scheduled times. By following these steps, we produce two corrected GTFS datasets used for generating median-corrected and dispersion-corrected accessibility measures. A flowchart of the entire process of calculating the P50 and

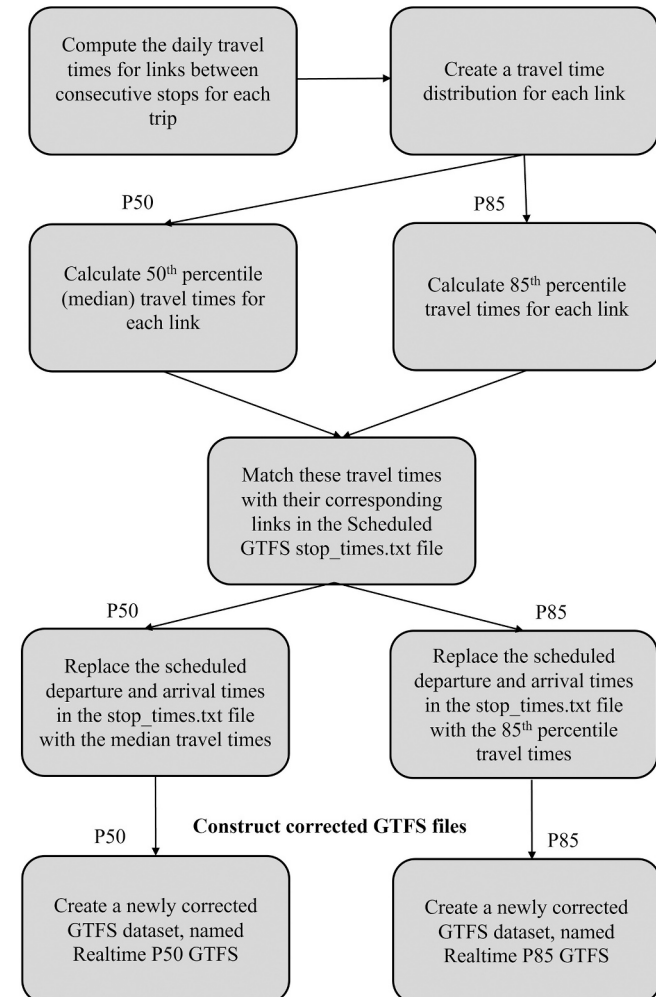


Fig. 1. A flowchart for creating Realtime P50 and P85 GTFS data.

P85 data is provided (Fig. 1).

To illustrate the key data processing steps for correcting Scheduled GTFS data, Fig. 2 presents a sample distribution of travel times for a transit link between Stop A and Stop B. This example assumes 100 weekdays of data, with each travel time observation representing the recorded duration of a transit trip along this link on a given weekday. The first step in our methodology is to construct the travel time distribution for each transit link. To account for travel time uncertainty, we calculate two key measures: the median travel time (50th percentile), which represents the central tendency of travel times and captures a typical trip duration, and the 85th percentile travel time, which provides a conservative estimate by accounting for variability. In Fig. 2, the green dashed line represents the median travel time, while the red dashed line marks the 85th percentile travel time. The next step is to integrate these refined travel time estimates into the Scheduled GTFS dataset to generate corrected versions that better reflect real-world transit conditions. As depicted in Fig. 1, this process involves correcting Scheduled GTFS travel times with median (50th percentile) travel times to generate the Realtime P50 GTFS dataset and using 85th percentile travel times to generate the Realtime P85 GTFS dataset.

2.2. Computing median- and dispersion-corrected accessibility

Using the Realtime P50 and P85 datasets median- and dispersion-corrected accessibility measures (Braga et al. 2023) are computed. First, we calculate travel times between each pair of origins (e.g., centroids of neighbourhoods) and destinations (e.g., job, healthcare, food). These travel times are then integrated into the cumulative opportunity method for measuring accessibility. The cumulative opportunity method counts the number of resources and opportunities within specific travel time isochrones—regions accessible within a set travel time threshold (Boisjoly and El-Geneidy, 2017; El-Geneidy and Levinson, 2006). The cumulative opportunity accessibility is computed as follows:

$$A_i^{Cumulative} = \sum_j f(t_{ij}) \quad (1)$$

where $A_i^{Cumulative}$ represents the cumulative-opportunity accessibility at the origin location i . The variable t_{ij} denotes the travel time between the origin i and the destination j . In this study, t_{ij} is obtained by summing the corrected travel times for each transit link along the path, using either the median (P50) or 85th percentile (P85) travel times from Realtime GTFS data.

The function $f(t_{ij})$ equals zero when the travel time between the origin i and the destination j exceeds the travel time threshold (e.g., 30 min, 1 h), and takes the value of one otherwise, as expressed below:

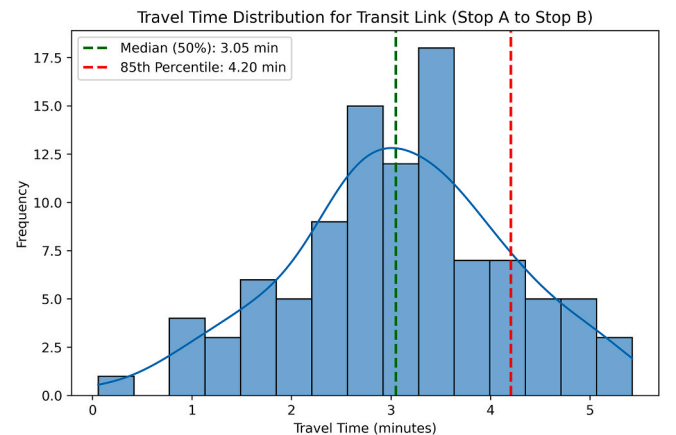


Fig. 2. Travel time distribution for a sample transit link between two consecutive stops.

$$f(t_{ij}) = \begin{cases} 1, & \text{if } t_{ij} \leq \text{threshold} \\ 0, & \text{if } t_{ij} > \text{threshold} \end{cases} \quad (2)$$

This equation uses Realtime P50 GTFS data for travel time calculations to derive the median-corrected accessibility measure. Conversely, using Realtime P85 GTFS data results in the dispersion-corrected accessibility measure.

Note that these median-corrected and dispersion-corrected accessibility measures address two main travel time uncertainty issues: 1) *inaccuracy* and 2) *variability* (Braga et al., 2023). Inaccuracy refers to discrepancies between the scheduled and actual travel times experienced by passengers (Braga et al., 2023). On the other hand, variability refers to the day-to-day fluctuations in travel times due to factors such as delays, congestion, accidents, signal control failures, weather conditions, service interruptions, and bus bunching (Braga et al., 2023; Chen et al., 2013; Lee and Kim, 2023; Liu et al., 2022b; Palm et al., 2020; Park et al., 2020). These variabilities can extend travel times from day to day even when passengers travel the exact same route at the exact same time leading to missed transit connections and delayed arrivals. Such variability is particularly problematic for socioeconomically vulnerable riders relying on transit for essential needs like healthcare (Lee and Kim, 2023; Park et al., 2020).

Recognizing the critical impact of travel time variability, several studies have proposed methods to incorporate it into accessibility analyses. For instance, Chen et al. (2013) modeled travel time variability using stochastic processes, while Palm et al. (2020) analyzed day-to-day delays to identify patterns in transit reliability through a social equity lens. Braga et al. (2023) used percentile-based measures, based on the 85th percentile travel time, to capture the upper bounds of variability, providing a more realistic assessment of transit system performance. These approaches highlight the importance of addressing variability not only to improve accessibility measures but also to support equitable and reliable transportation planning.

Table 1 compares the three accessibility measures used in this study, focusing on their ability to address two key challenges to travel time uncertainty: inaccuracy and variability. Scheduled accessibility neglects both travel time inaccuracy and variability present in the Scheduled GTFS data. Median-corrected accessibility accounts for travel time inaccuracy but does not address variability (Braga et al., 2023). In contrast, dispersion-corrected accessibility considers both travel time inaccuracy and variability.

Table 1
Comparison of three accessibility measures in relation to their consideration of travel time inaccuracy and day-to-day travel time variability.

	Accounts for inaccuracy in the Scheduled GTFS data	Accounts for day-to-day travel time variability	Description
Scheduled accessibility	No	No	Relies on Scheduled GTFS data, assumes perfect adherence to schedules, and ignores real-world conditions
Median-corrected accessibility	Yes	No	Corrects for inaccuracy by incorporating median travel times from Realtime GTFS data but does not account for variability
Dispersion-corrected accessibility	Yes	Yes	Corrects for both inaccuracy and variability by incorporating the 85th percentile travel times from Realtime GTFS data

3. Case study: examining the impacts of travel time uncertainty on transit project evaluation

3.1. Overview

To showcase the applicability of the framework, we design and implement an empirical study. This case study explores how ignoring two main travel time uncertainty issues: inaccuracy and variability, might distort the assessment of a new public transit service's impacts on healthcare accessibility and inequalities. Specifically, we compare:

- Scheduled accessibility measures (based on Scheduled GTFS), which do not consider travel time inaccuracy and variability issues,
- Median-corrected accessibility measures (based on Realtime P50 GTFS), which account for travel time inaccuracy but overlook variability issues, and
- Dispersion-corrected accessibility measures (based on Realtime P85 GTFS), which address both travel time inaccuracy and variability issues.

By comparing the results obtained from these different accessibility measures, we illustrate the impacts of travel time inaccuracy and variability issues on transit project evaluation. In particular, to assess the impacts of travel time inaccuracy, we compare scheduled accessibility with median-corrected accessibility. For examining the impacts of travel time variability, we compare median-corrected accessibility and dispersion-corrected accessibility. To illustrate the combined impacts of travel time inaccuracy and variability, we compare scheduled accessibility with dispersion-corrected accessibility.

We conduct this comparative study using the city of Columbus, Ohio, USA, as our case study area. The northeastern part of Columbus is characterized by economically disadvantaged communities that have faced significant health disparities, including high rates of premature births and obstacles to healthcare accessibility (Laura Bliss, 2016; Root et al., 2020). To address these issues, the city introduced the CMAX bus rapid transit (BRT) system in 2018, operated by the Central Ohio Transit Authority (COTA) (Laura Bliss, 2016). We use the CMAX bus rapid transit (BRT) system as a new transit service in this example study. The CMAX BRT service, depicted by the red line in Fig. 3(a), complements the existing transit network shown by the blue lines. To evaluate the localized impacts of the BRT service on accessibility and inequality, we consider a 524-meter transit service area and select block groups located within this designated service area for the analysis (El-Genaidy et al., 2014).

3.2. Data

We obtain the Scheduled GTFS data from OpenMobilityData (2024), which provides a standard format for representing public transit schedules and routes (Google, 2024b). For our analysis, we select the first Tuesday of December for both 2017 and 2018, specifically December 4, 2017, and December 3, 2018. The GTFS data spans from September 4 to December 31, 2017, and from September 3, 2018, to January 6, 2019. This allows for direct comparisons before and after the introduction of the BRT service.

To estimate empirical travel time distributions for each link within the transit (i.e., the travel time between consecutive stops), we employ the Realtime GTFS data from the local transit agency. This includes trip updates, service alerts, and vehicle spatial locations (Google, 2024a). Consistency in the data collection period is crucial for accurately matching the trip update information from Realtime GTFS data with Scheduled GTFS data. Therefore, we collect Realtime GTFS data for December 2017 and 2018 from the official application programming interface (API) of the Central Ohio Transit Authority (COTA). This data is stored in a MongoDB database for subsequent processing and analysis using Python. Table 2 summarizes the versions of Scheduled and

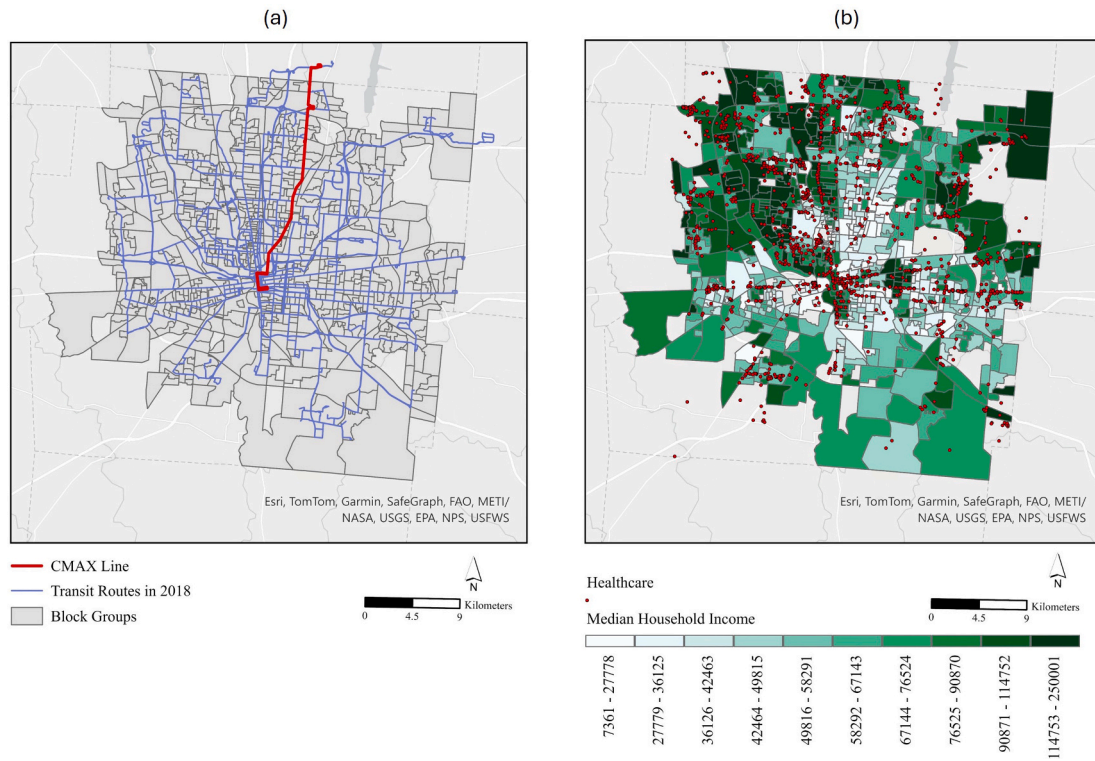


Fig. 3. Maps of a) the transit network in 2018 and the CMAX line and b) the geographic distribution of median household income and healthcare facilities across block groups in Columbus.

Table 2

Scheduled and Realtime GTFS data used in this study.

	Scheduled GTFS		Realtime GTFS
	Publication date	Analysis date	
Pre CMAX	Published on September 4, 2017	December 4, 2017	December 1–31, 2017
Post CMAX	Published on September 3, 2018	December 3, 2018	December 1–31, 2018

Realtime GTFS data utilized in this study.

3.3. Analyses

3.3.1. Generating Realtime P50 and P85 GTFS

The Realtime GTFS trip updates for December 2017 and 2018 were provided in BSON and JSON formats and imported into MongoDB using Studio3T to enhance data management efficiency. In MongoDB, the data is organized into collections (equivalent to tables), with each collection corresponding to a single day. Each document within these collections contains critical information, including Stop Sequence (denoting the sequence of stops for a specific trip), Timestamp (indicating the last update time in POSIX format), trip ID (identifying a unique instance of a vehicle's journey along a route at a specified time), route ID; and Vehicle ID. This structured approach ensures systematic storage and retrieval of transit data within the MongoDB database (Google, 2024a).

Using Python, we organize the data by creating a unique database for each trip ID specific to each year. Within each database, we establish multiple collections, each representing a different day in December. For instance, the database for trip ID 604997 in 2017 contains 29 collections, one for each day of the month. Each collection includes details such as stop name, updated arrival time, and document ID. We compute the travel time between each consecutive stop for each trip within every collection. We focus exclusively on the weekdays in December, a total of

21 days, which corresponds to the typical operational days of most healthcare facilities. From this data, we construct travel time distributions and calculate both the median and the 85th percentile travel times between consecutive stops for each trip across all selected weekdays.

With the median and 85th percentile travel times calculated for each link, we match these travel times with their counterparts in the Scheduled GTFS, specifically within the *stop_times.txt* file. We correct the departure and arrival times at each stop based on the median travel times and create a corrected version of the dataset named Realtime P50 GTFS for Columbus. Similarly, we apply the 85th percentile travel times to the corresponding links in the Scheduled GTFS, which results in another corrected dataset called Realtime P85 GTFS as shown in Fig. 1. It is noteworthy that we assume that the initial departure times provided in the Scheduled GTFS feed are accurate. To promote reproducibility and replicability, we provide a [link](#) to the full set of scripts for data processing and analysis in Python and MongoDB.

3.3.2. Accessibility analysis

Having obtained the Scheduled GTFS data, along with Realtime P50 GTFS and Realtime P85 GTFS, we calculate transit travel times between multiple pairs of origins (centroids of block groups) and destinations (healthcare locations) for specific departure times and time windows to measure accessibility to healthcare. The healthcare data, sourced from the DataAxle business dataset, encompasses a range of facilities such as hospitals, clinics, and various healthcare practitioners (see Fig. 3(b)) (Data Axle, 2023).

We use *r5r* (Pereira et al., 2021), an open-source R package known for rapid realistic routing, to construct a routable transport network. This tool allows us to measure minute-by-minute transit travel times across our multimodal transportation network using the *travel_time_matrix()* function. We specifically set 6:30 AM as our departure time, with a 150-minute travel time window extending until 9:00 AM, aligning with morning peak hours defined by the local transit agency (Central Ohio Transit Authority, 2024). This travel time window allows us to capture the fluctuations in transit schedules during morning peak

periods, with the *r5r* computing travel times for each minute within this interval (Saraiva et al., 2024). The median travel time is then used as the standard measure for assessing accessibility between each origin–destination pair (Conway et al., 2018). Given that the typical walking distance to transit stops ranges from 328 to 520 m (Daniels and Mulley, 2013; El-Geneidy et al., 2014; Tennøy et al., 2022), we set the maximum walking distance of 500 m for our travel time matrix calculations.

Using the *r5r* tool, we create two travel time matrices for each of the three GTFS datasets—Scheduled, Realtime P50, and Realtime P85—using December 4, 2017, and December 3, 2018, as departure dates. This results in a total of six travel time matrices, which contain transit travel times between pairs of origins (centroids of block groups) and destinations (healthcare locations). Each GTFS dataset provides a distinct set of travel times that we use to generate three different cumulative-opportunity accessibility measures (Eqs. (1) and (2)) with a 60-minute travel time threshold, a commonly used threshold for assessing regional-level healthcare accessibility (Lee and Miller, 2018; Singh et al., 2022): 1) scheduled accessibility (based on the Scheduled GTFS data); 2) median-corrected accessibility (based on the Realtime P50 GTFS data); and 3) dispersion-corrected accessibility (based on the Realtime P85 GTFS data).

3.3.3. Inequality assessment

In this section, we describe our two approaches to evaluating accessibility inequalities. First, we assess whether accessibility changes were unevenly distributed across block groups with different socioeconomic characteristics. Specifically, we classify block groups into two categories: those that experienced increased accessibility and those that experienced decreased accessibility following the BRT implementation. We then compare these two groups across four socio-economic indicators: 1) unemployment rate, 2) low education rate, 3) public transit usage rate, and 4) non-white population ratio. Given that these variables do not follow a normal distribution, the Mann-Whitney *U* test provides a non-parametric and robust method for assessing whether one group systematically exhibits higher or lower values for each indicator. In this analysis, low education level is defined as the percentage of the population with less than a high school diploma, using data from the educational attainment of the population 25 years and over dataset. The unemployment rate is measured based on the civilian population in the labor force aged 16 and over, while the public transit usage rate is derived from the means of transportation to work for workers aged 16 and over dataset. All socio-economic indicators are obtained from the 2016–2020 American Community Survey (ACS) five-year estimates (U.S. Census Bureau, 2020). By applying this statistical approach, we aim to determine whether there are statistically significant differences in socio-economic variables between neighbourhoods with increased and decreased accessibility before and after the new BRT service.

Next, we employ the Palma ratio, an easy-to-communicate measure of inequality introduced by Palma (2011). This ratio compares the income share of the top 10 % of earners to that of the bottom 40 %, which are the income distribution segments most prone to change (Karner and Pereira, 2024). In transportation studies, the Palma Ratio is frequently used to compare the average accessibility between the wealthiest 10 % of block groups and the poorest 40 % (Guzman and Oviedo, 2018; Herszenhut et al., 2022; Javanmard et al., 2024; Liu et al., 2022a; Shaver et al., 2025). This adaptation allows for the examination of socio-economic differences in spatial accessibility distribution. A Palma ratio exceeding 1 indicates that more affluent block groups experience greater accessibility (or accessibility gains from the new transit service) compared to less affluent block groups. Conversely, a ratio below 1 suggests that the less affluent block groups have higher accessibility (or accessibility gains from the new transit service) compared to the wealthier census block groups (Tomasiello et al., 2022). We obtain and use median household income data from the 2016–2020 American Community Survey five-year estimates (U.S. Census Bureau, 2020) for the calculations of Palma ratios. We apply this Palma ratio to examine

inequalities in accessibility values themselves as well as positive accessibility changes (i.e., gains) before and after the BRT construction. It is noteworthy that, since the Palma ratio is defined for positive values, our analysis focuses exclusively on accessibility gains, rather than losses. Understanding how these accessibility gains were distributed across income groups is particularly important, given that one of the primary goals of this BRT implementation was to improve transit accessibility for low-income populations, particularly in historically underserved neighbourhoods such as Linden (Laura Bliss, 2016; Lee and Miller, 2018). Since the core objective of this transit investment was to enhance mobility for lower-income communities, it is critical to assess whether the accessibility benefits were equitably distributed or disproportionately concentrated among higher-income groups. By applying the Palma ratio, we can evaluate the extent to which this transit improvement effectively served its intended beneficiaries, providing insights into whether it contributed to reducing or reinforcing existing accessibility inequalities.

3.4. Results

3.4.1. Impacts of travel time uncertainty on healthcare accessibility and inequality

Before examining the impacts of travel time inaccuracy and variability on the evaluation of the transit project, it is valuable to understand how accessibility and its inequality might change under such travel time uncertainty issues. To establish a baseline for the analysis, we first examine accessibility levels before the BRT implementation.

The maps of transit accessibility (Fig. 4) reveal a consistent pattern where central block groups exhibit higher accessibility to healthcare compared to peripheral block groups. This disparity in accessibility levels could be attributed to the denser transportation network, higher transit service frequency, and the concentration of healthcare facilities in the city's core (see Fig. 3(b)). The scheduled accessibility (based on Scheduled GTFS) is generally higher than both median-corrected accessibility (based on Realtime P50 GTFS) and dispersion-corrected accessibility (based on Realtime P85 GTFS). However, distinguishing the differences between the maps in Fig. 4(a)–(c) is challenging and offers limited insight into the impacts of travel time inaccuracy on accessibility measurements. Therefore, to better understand the impacts of travel time uncertainty on accessibility values and to clearly illustrate the differences between each pair of maps, we calculate the accessibility percentage change $((y - x)/x) \times 100$ (see Fig. 4).

First, we compare scheduled accessibility (Fig. 4(a)) against median-corrected accessibility (Fig. 4(b)) to examine the impacts of travel time inaccuracy issue. Here, *y* represents median-corrected accessibility and *x* signifies scheduled accessibility. The prevalence of red block groups suggests that scheduled accessibility measures generally overestimate accessibility relative to median-corrected accessibility metrics, underscoring the significance of addressing travel time inaccuracy. Specifically, the average accessibility percentage change between scheduled and median-corrected accessibility is -5.91% (see Fig. 5). Furthermore, the prevalence of red block groups in the central regions highlights the potential disproportionate effects of travel time inaccuracy on accessibility measurements in these areas. This overestimation is particularly pronounced in central neighbourhoods, where congested roads and potential limitations in the quality of the Scheduled GTFS data could be significant factors (Braga et al., 2023). On the contrary, in peripheral regions farther from transit services, the impacts of travel time inaccuracy on accessibility levels appear less pronounced. Several block groups shown in blue indicate cases where scheduled accessibility is less than median-corrected accessibility. This pattern aligns with the findings of (Wessel and Widener, 2017), who noted that median-corrected accessibility is often higher than scheduled accessibility. This discrepancy typically arises from the conservative scheduling approaches that anticipate delays that may not occur in practice (Wessel and Widener, 2017). The overestimation of scheduled accessibility compared to



Fig. 4. Spatial patterns of (a) scheduled accessibility, (b) median-corrected accessibility, and (c) dispersion-corrected accessibility, along with accessibility percentage changes between: (d) scheduled and median-corrected accessibility, (e) median and dispersion-corrected accessibility, and (f) scheduled and dispersion-corrected accessibility. Percentage change is calculated as $[(\text{Measure 2} - \text{Measure 1}) / \text{Measure 1}] \times 100$.

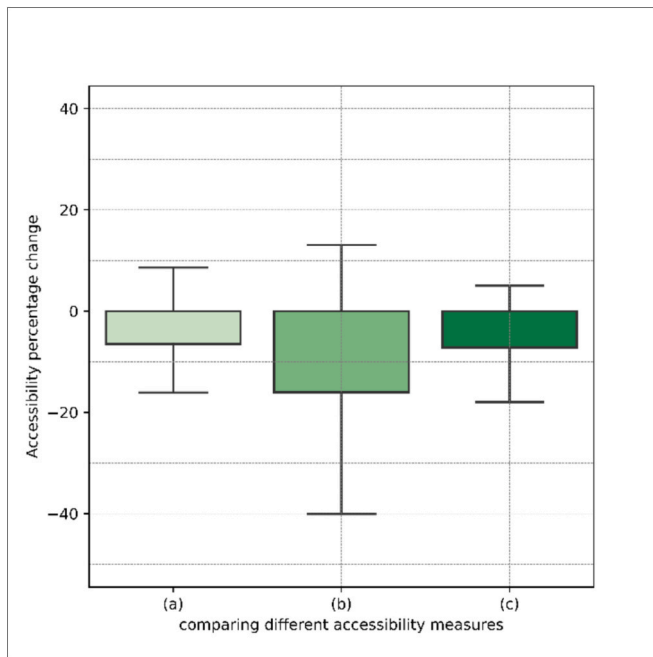


Fig. 5. Box plots for accessibility percentage changes between (a) scheduled and median-corrected accessibility, (b) scheduled and percentile-corrected accessibility, and (c) median-corrected and dispersion-corrected accessibility.

median-corrected accessibility highlights the importance of incorporating travel time inaccuracy issues to provide more accurate accessibility measurements.

Secondly, we explore the impacts of travel time variability by comparing median-corrected accessibility (Fig. 4(b)) and dispersion-corrected accessibility (Fig. 4(c)). In this comparison, x represents median-corrected accessibility, while y refers to dispersion-corrected accessibility. The analysis reveals that median-corrected accessibility surpasses dispersion-corrected accessibility, with an average overestimation of 4.81 % (see Fig. 5). This trend is indicated by the predominance of red block groups on the map. In contrast, certain areas reflect a scenario where median-corrected accessibility is less than dispersion-corrected accessibility, denoted by blue block groups. This indicates that median-corrected accessibility measures using median-corrected datasets may not fully capture the variability in travel times.

Lastly, we assess the differences between scheduled accessibility and dispersion-corrected accessibility to evaluate the combined impacts of both travel time inaccuracy and variability. When we calculate the accessibility percentage change here, x is scheduled accessibility, and y is dispersion-corrected accessibility. The analysis shows a majority of block groups colored in red. This indicates an overestimation by scheduled accessibility measure relative to dispersion-corrected accessibility measure. The average accessibility percentage change of -10.97 % underscores the notable impacts of travel time inaccuracy and variability on accessibility metrics (see Fig. 5). The overestimation by scheduled accessibility underscores the need to account for travel time uncertainty issues such as inaccuracy and variability in planning and accessibility measurements.

To explore the potential impacts of travel time uncertainty issues on inequality assessment, we calculate and compare Palma ratios using three different accessibility measures. The calculated Palma ratios are 0.29 for scheduled accessibility, 0.26 for median-corrected accessibility, and 0.24 for dispersion-corrected accessibility (see Table 3). Lower Palma ratios indicate relatively better accessibility to healthcare for low-income communities compared to high-income counterparts. Despite the differences in Palma ratios, all consistently remain below one. This highlights that lower-income block groups generally have higher

Table 3

Palma ratios based on different accessibility measures.

Accessibility measure	Palma ratio
Scheduled accessibility	0.29
Median-corrected accessibility	0.26
Dispersion-corrected accessibility	0.24

healthcare accessibility than wealthier block groups, regardless of the data input.

To determine the statistical significance of differences in Palma ratios, we employ the bootstrapping approach (Javanmard et al., 2024). Results show that the differences between Palma ratios calculated using the three different accessibility measures are not statistically significant at the threshold of p -value 0.05. Consequently, travel time uncertainty does not substantially influence the evaluation of inequality in the distribution of healthcare accessibility.

3.4.2. Impacts of travel time uncertainty on transit project evaluation

In this section, we explain how neglecting travel time uncertainty issues, namely inaccuracy and variability, can lead to an erroneous evaluation of new transit services. We calculate the percentage changes in healthcare accessibility before and after the introduction of the BRT service, using scheduled, median-corrected, and dispersion-corrected accessibility measures. Moreover, we use the Mann-Whitney U test to examine whether socio-economic variables significantly differ between census block groups that experienced increased versus decreased accessibility. Finally, we investigate the inequalities in accessibility gains across block groups with different incomes by using Palma ratios.

Fig. 6 illustrates the percentage changes in accessibility to healthcare before and after BRT implementation. Block groups in red reflect a decrease in accessibility exceeding 10 %, while those in blue indicate an increase greater than 10 %. In Fig. 6(a), where the scheduled accessibility measure is used without accounting for inaccuracy and variability, a mixed pattern of positive and negative impacts of the new transit service on healthcare accessibility is observed. However, Fig. 6(b) and (c), where median-corrected and dispersion-corrected accessibility measures are used respectively, demonstrate more distinct positive impacts, especially in block groups near the transit line in the city's central area.

The average percentage change in accessibility is around 5 % with the scheduled accessibility measure, which contrasts sharply with the 57 % and 42 % increases observed with median-corrected and dispersion-corrected measures, respectively. These discrepancies underscore the risks of depending on scheduled measures, which fail to account for travel time uncertainty issues when evaluating new transit services. Such reliance can result in assessments that misjudge the true effects of the BRT service on healthcare accessibility.

Notably, the greater benefits of CMAX service observed with median-corrected and dispersion-corrected accessibility measures can be attributed to transit infrastructure enhancements such as dedicated BRT lanes and Transit Signal Priority (TSP) systems. TSP systems activate green lights as buses approach intersections. This helps to mitigate signal and traffic delays, ultimately improving overall transit service reliability and efficiency (Central Ohio Transit Authority, 2024).

Tables 4–6 present the Mann-Whitney U test results for three different accessibility measures: scheduled accessibility (Table 4), median-corrected accessibility (Table 5), and dispersion-corrected accessibility (Table 6). Across all three measures, the results indicate no statistically significant differences at a 5% significance level in socio-economic variables between block groups with increased and decreased accessibility. This suggests that the new BRT service did not appear to systematically favor or disadvantage block groups based on their racial composition, education, unemployment levels, or transit dependency, though further research may be needed to explore potential disparities. Building on these findings, the next section focuses exclusively on

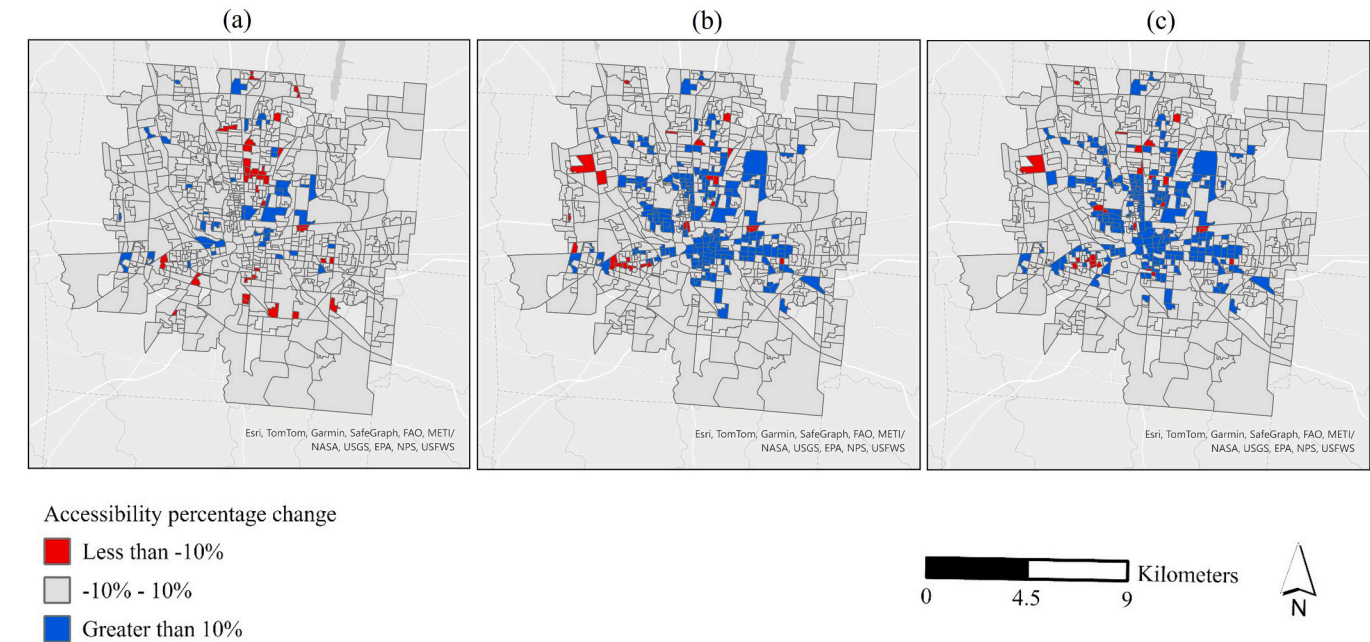


Fig. 6. Spatial patterns of accessibility percentage change using (a) scheduled accessibility, (b) median-corrected accessibility, and (c) dispersion-corrected accessibility measures.

Table 4
Comparison of socio-economic characteristics between block groups with increased and decreased accessibility based on the scheduled accessibility measure.

Variables	Median		P-values
	Block groups with increased accessibility	Block groups with decreased accessibility	
Unemployment Rate (%)	4.83	6.08	0.79
Low Education Rate (%)	8.68	15.24	0.06
Public Transit Usage Rate (%)	1.51	3.66	0.63
Non-White Population Ratio (%)	43.59	40.26	0.82

Table 5
Comparison of socio-economic characteristics between block groups with increased and decreased accessibility based on the median-corrected accessibility measure.

Variables	Median		P-values
	Block groups with increased accessibility	Block groups with decreased accessibility	
Unemployment Rate (%)	5.06	3.49	0.32
Low Education Rate (%)	7.29	11.99	0.06
Public Transit Usage Rate (%)	2.74	2.73	0.20
Non-White Population Ratio (%)	35.88	38.21	0.65

accessibility *gains*, examining whether the benefits of improved transit access were equitably distributed across income groups using the Palma ratio.

Table 6
Comparison of socio-economic characteristics between block groups with increased and decreased accessibility based on the dispersion-corrected accessibility measure.

Variables	Median		P-values
	Block groups with increased accessibility	Block groups with decreased accessibility	
Unemployment Rate (%)	5.03	4.60	0.16
Low Education Rate (%)	7.48	13.96	0.12
Public Transit Usage Rate (%)	2.17	3.17	0.06
Non-White Population Ratio (%)	36.03	37.20	0.97

To explore the distribution of accessibility benefits (i.e., gains) of the new transit service across different income groups, we employ Palma ratios, which compare the average accessibility gains received by the bottom 40 % of block groups against those experienced by the top 10 %. The Palma ratios using scheduled, median-corrected, and dispersion-corrected accessibility measures are approximately 0.06, 0.25, and 0.32, respectively, all falling below one (see Table 7). These ratios suggest an unequal distribution of accessibility benefits across all accessibility measures, with lower-income block groups receiving greater transit accessibility gains compared to higher-income block groups. This outcome aligns with the intended objective of the new transit service (Laura Bliss, 2016; OHM, 2024).

Table 7
Palma ratios on accessibility gains based on different accessibility measures.

GTFS data used	Palma ratio
Scheduled accessibility	0.06
Median-corrected accessibility	0.25
Dispersion-corrected accessibility	0.32

However, the Palma ratio derived from scheduled accessibility is notably lower than those from median-corrected and dispersion-corrected accessibility, as shown in Table 7. Using the bootstrapping approach, we find that the difference between the Palma ratio from scheduled accessibility (0.06) and that from dispersion-corrected accessibility (0.32) is statistically significant (see Table 8). This statistically significant difference suggests that scheduled accessibility measures may not accurately reflect the empirical benefits of the new transit service, while dispersion-corrected accessibility measures demonstrate the positive impacts of new transit services on reducing accessibility inequalities. These differences underscore the distortions that can arise from neglecting travel time inaccuracy and variability in scheduled accessibility measures, potentially resulting in misleading assessments of the impacts of transit interventions on accessibility inequality.

4. Discussion and conclusion

This study presents a practical framework that utilizes a standardized format for real-time transit data: Realtime GTFS data, to compute easy-to-use transit accessibility measures under travel time uncertainty. The framework begins by generating two corrected Scheduled GTFS datasets: Realtime P50 GTFS and Realtime P85 GTFS. These datasets are then employed to develop two cumulative-opportunity measures that account for travel time uncertainty: median-corrected accessibility (using Realtime P50 GTFS) and dispersion-corrected accessibility (using Realtime P85 GTFS) (Braga et al., 2023). By incorporating travel time uncertainty, these measures provide more accurate accessibility assessments, reflecting the actual experience of transit users better than the scheduled accessibility measure. Through an empirical case study in Columbus, Ohio, we then evaluate the impacts of travel time inaccuracy and variability issues on healthcare accessibility, inequality, and transit project evaluation.

The findings of our example study in Columbus reveal that scheduled accessibility, which ignores both inaccuracy and variability, overestimates accessibility. This overestimation can lead to misleading conclusions about the accessibility benefits of new transit services. Specifically, the evaluation of the transit project shows that scheduled accessibility fails to capture the benefits of the new transit service on healthcare accessibility. However, when travel time uncertainty is taken into account by using median-corrected and dispersion-corrected accessibility measures, the positive impacts of the new transit service on healthcare accessibility become evident. This implies that ignoring travel time uncertainty can obscure the benefits of transit services on accessibility.

Despite differences in accessibility values themselves (Fig. 4) calculated using scheduled, median-corrected, and dispersion-corrected measures, our results indicate that these variations do not impact the overall assessment of inequality in healthcare accessibility (Table 3). Across all measures, the Palma ratios consistently remain below one, highlighting that lower-income block groups generally have greater healthcare accessibility than wealthier block groups. Similarly, despite

differences in accessibility gains (Fig. 6) computed using the three measures, the overall conclusion of the inequality analysis remains unchanged. As shown in Table 7, all Palma ratios fall below one, indicating an unequal distribution of accessibility benefits across all measures, with lower-income block groups experiencing greater transit accessibility gains than higher-income block groups.

This study acknowledges several limitations that suggest directions for future research. Firstly, we used a single travel time threshold of 60 min in our cumulative opportunity approach for measuring accessibility. This can be expanded in future studies to explore how different thresholds influence findings (Pereira, 2019). It is also important to note that the 85th percentile travel times were chosen for generating dispersion-corrected accessibility measures to capture significant delays while avoiding an overemphasis on extreme outliers that may not reflect typical daily transit conditions, following Braga et al. (2023). Future research could explore alternative thresholds, such as the 95th percentile, to assess their impacts on accessibility and inequality measures. In our study, the travel time between an origin and a destination is computed by summing the corrected travel times for each transit link along the path, using either the median (P50) or 85th percentile (P85) travel times from Realtime GTFS data (Braga et al., 2023). While this method offers a computationally efficient and easy-to-implement way to estimate travel times under uncertainty, it assumes that percentile-based travel times at different links occur simultaneously. More advanced stochastic path-based simulations could further refine these estimates by dynamically modeling real-time variability across links in a way that accounts for their interdependencies, ensuring that fluctuations in travel time propagate realistically along the transit path. Nevertheless, our approach remains valuable for its simplicity and applicability in transit accessibility analyses. Further, previous research has highlighted the significance of supply-demand dynamics and distance decay in healthcare accessibility measurements and transit project appraisals (Javanmard et al., 2024). Future studies could develop new measures that incorporate travel time uncertainty, using advanced accessibility methodologies such as the Multimodal Two-Step Floating Catchment Area (2SFCA) and the Multimodal Generalized Two-Step Floating Catchment Area (G2SFCA) as the basis to further validate our conclusions. Additionally, our analysis was constrained by data availability, particularly in the pre-post BRT analysis, which was limited to December. However, travel times—and consequently, accessibility levels—are subject to seasonal and weekly fluctuations influenced by factors such as weather conditions and traffic patterns. Investigating these seasonal variations, especially in relation to travel time uncertainty, represents a promising direction for future research. Also, future studies could enhance accessibility inequality analysis by incorporating socio-demographic factors such as age, gender, and race using spatial regression models. This approach would provide a more comprehensive evaluation of how new transit services affect accessibility inequalities across different demographic groups (Lee and Kim, 2023). Another limitation of this study is that it considers only buses as the primary mode of transportation in our case study in Columbus. While the framework we developed is adaptable to multimodal transit systems—such as those in cities like New York City, which include buses, subways, and other modes—the real-time data collection process becomes more complex. Applying this framework in such cities requires integrating real-time data from multiple modes, posing challenges in data accuracy and consistency. Future research could extend this framework to cities with multimodal transport systems, examining how multimodal data influences the assessment of transit accessibility and inequality. Also, it is important to recognize that the impacts of travel time uncertainty on transit project evaluation may vary between cities due to differences in methods used by transit agencies to develop their scheduled GTFS data, as well as variations in data quality, land use patterns, traffic volumes, and weather conditions. To address this variability, we recommend that transit planners undertake similar analyses tailored to their respective cities. Lastly, it is also important to note that

Table 8
Differences in Palma ratios based on different accessibility measures with statistical significance indicated.

	Scheduled accessibility	Median-corrected accessibility	Dispersion-corrected accessibility
Scheduled accessibility	0	-0.19	-0.26
Median-corrected accessibility		0	-0.08
Dispersion-corrected accessibility			0

Note: Numbers in bold indicate statistical significance at the 0.05 level.

Realtime GTFS data is not widely available in small- to medium-sized cities or low- and middle-income countries (LMICs). In these regions, even Scheduled GTFS data is often scarce, posing challenges for accurate transit-based accessibility analysis (Kim et al., 2023; Williams et al., 2015). To address this, cost-effective methods to collect and generate Realtime GTFS data must be explored. For example, in Ulaanbaatar, a scheduled GTFS dataset was created using open-source tools like the U. S. National Rural Transit Assistance Program (RTAP) GTFS Builder, which can be adapted for resource-limited regions (Kim et al., 2023). Similarly, the Digital Matatus project in Nairobi demonstrated how mobile phones could be used to map the city's semi-formal matatu bus system and convert the data into GTFS format for broader use (Williams et al., 2015). This approach utilized crowd-sourcing to collect data, which was then integrated into Google Maps, offering valuable insights for transit planning in Nairobi. For real-time GTFS data, similar low-cost approaches—such as GPS-enabled smartphones, sensors, or crowd-sourced data collection—could be explored. Equipping transit vehicles with GPS devices and developing simple platforms for real-time data sharing could enable even resource-constrained regions to establish basic real-time GTFS datasets. These efforts would enhance transit planning and enable transport geographers and GIScientists to conduct more precise analyses in underrepresented areas.

Despite these limitations, our study introduces a practical framework for correcting Scheduled GTFS data using Realtime GTFS data, resulting in more accurate transit accessibility measures under travel time uncertainty. By incorporating real-time data into transit project evaluations, particularly for BRT systems, our approach allows for more accurate assessments of their impacts, including accessibility benefits for low-income neighbourhoods. Traditional methods relying on scheduled accessibility measures tend to underestimate the benefits by ignoring travel time uncertainty, which can result in misleading conclusions. By using median-corrected and dispersion-corrected accessibility measures, our study demonstrates the notable positive impacts of the new BRT system on healthcare accessibility, especially for underserved communities.

These findings have important policy implications for transportation planning, especially for local Metropolitan Planning Organizations (MPOs) and Departments of Transportation (DOTs) responsible for evaluating and implementing transit projects. This study highlights the crucial role of travel time uncertainty in accurately assessing the benefits of new transit systems, such as BRT systems. Scheduled GTFS data often fail to account for real-world travel time variability, leading to an overestimation of accessibility measurements and an underestimation of transit benefits. Therefore, transportation planners should prioritize incorporating Realtime GTFS data into their evaluation processes to ensure a more accurate and fair assessment of transit benefits. This will help policymakers make informed decisions that fully reflect the impact of transit investments.

Furthermore, when advocating for new transit systems, relying solely on scheduled accessibility data may not fully capture the true value of improvements. Realtime GTFS data provides a clearer picture of the system's benefits, supporting the case for funding, maintenance, and expansion by accurately reflecting the system's impact. Adopting measures that account for travel time uncertainty will allow MPOs and DOTs to design more equitable transit systems that better serve underserved populations. By integrating more accurate and comprehensive data, planners can make better-informed decisions about where to invest in transit infrastructure, prioritizing areas for improving accessibility and reducing inequality.

CRediT authorship contribution statement

Reyhane Javanmard: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Luyu Liu:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Jed A.**

Long: Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Jinhyung Lee:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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